

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457394.69158 +/- 0.00023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.022
Transit depth (Rp/Rs)^2	1.91 +/- 0.61 [%]
Orbital Inclination (inc)	83.59 +/- 2.85
Airmass coefficient 1 (a1)	1.016 +/- 0.017
Airmass coefficient 2 (a2)	-0.009 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.66 %
Best Comparison Star	#1 - [289, 204]
Optimal Aperture	3.6
Optimal Annulus	9.19
Transit Duration (day)	0.1264 +/- 0.0062

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.138 ± 0.022 vs 0.1178 (deviation 0.83σ)
Duration fit vs published	0.1264 d vs 0.125 d (deviation 0.2σ)
Mid-time O-C	-1571.3 min
Depth SNR	5.7
Residual scatter	1.66 %

Light curve

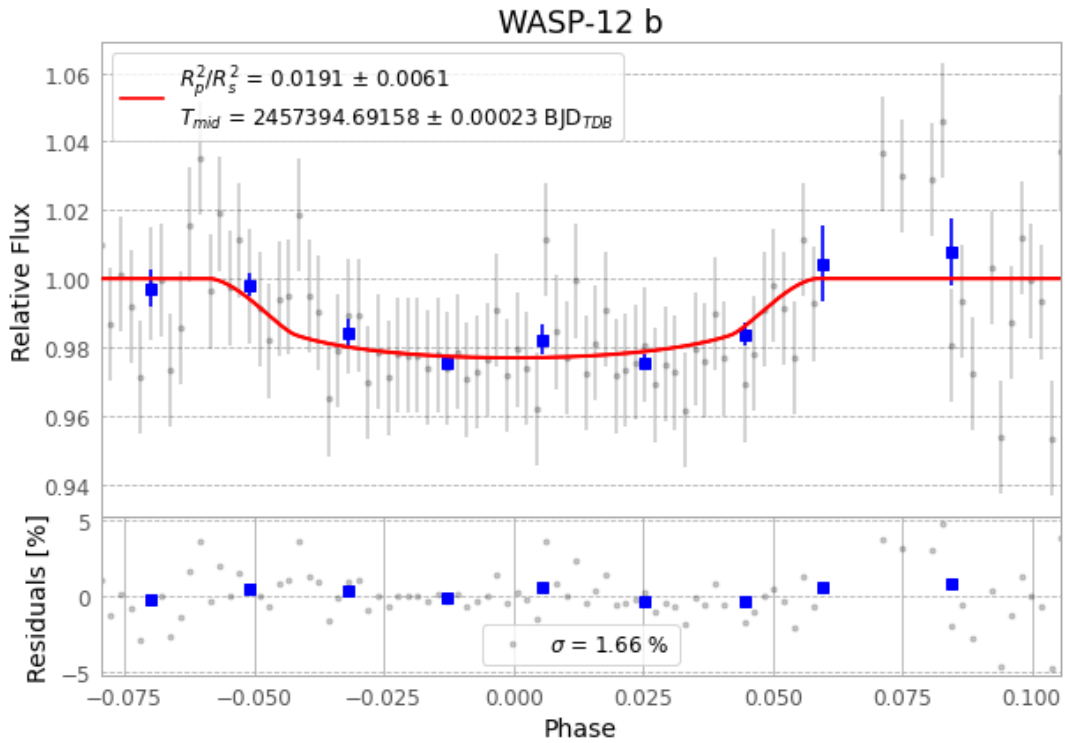


Figure 1 — FinalLightCurve_WASP-12 b_2016-01-06.png (SHA-256 ea5afd14d35adfb7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [367, 168], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 044a0a0be7fc4824...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

98 FITS frames (64 MB), WASP-12160107022112.FITS → WASP-12160107071212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-160108.log (882abb78555b...), FinalLightCurve_WASP-12 b_2016-01-06.png (ea5afd14d35a...), FinalLightCurve_WASP-12 b_2016-01-06.csv (d72c14be87f6...)

Compute resources

Wall time 596.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 13:42Z.